

Minor syllabus

THE MAHATMA GANDHI UNIVERSITY

UNDERGRADUATE VOCATIONAL

PROGRAMMES

(HONOURS) SYLLABUS

MGU-B.VOC. (Honours)

(2025 Admission Onwards)



Faculty: Science

Expert Committee: Core Industrial and infra

Structure Skills

Subject: Environmental Science

Mahatma Gandhi University

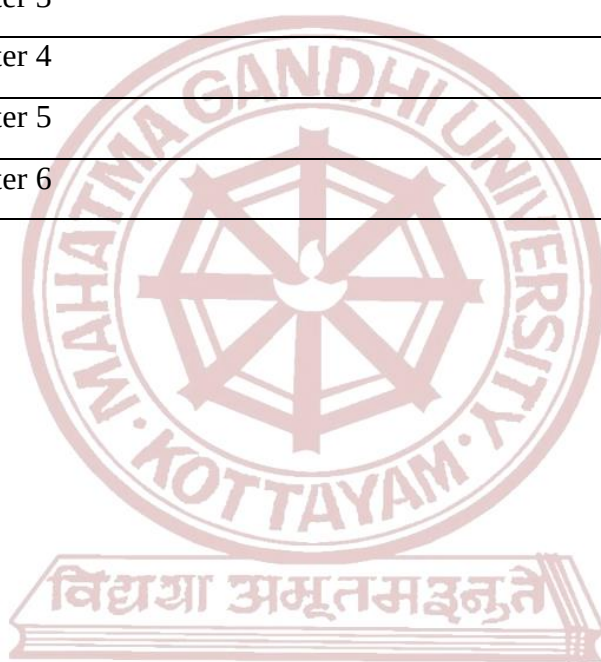
Priyadarshini Hills

Kottayam – 686560, Kerala,

Syllabus Index

Contents

SL.NO.	TITLE	PAGE NO.
1	Syllabus Index	3 – 4
2	Semester 1	5 – 11
3	Semester 2	12 – 17
4	Semester 3	18 – 23
5	Semester 4	24 – 29
6	Semester 5	30 – 34
7	Semester 6	35 - 39



MGU-B.VOC. (HONOURS)

Syllabus

Syllabus Index

Name of the Minor: Environmental Science

Semester: 1

Course Code	Title of the Course	Type of the Course	Credit	Hours/ week	Hour Distribution /week		
					L	P	O
MG1MPCESM100	Environment and Ecology	MPC	4	5	3	2	0

L — Lecture, P — Practical/Practicum , O — On the Job Training

Semester: 2

Course Code	Title of the Course	Type of the Course	Credit	Hours/ week	Hour Distribution /week		
					L	P	O
MG2MPCESM100	Environmental Education and Biodiversity Conservation	MPC	4	4	4	0	0

L — Lecture, P — Practical/Practicum , O — On the Job Training

Semester: 3

Course Code	Title of the Course	Type of the Course	Credit	Hours/ week	Hour Distribution /week		
					L	P	O
MG3MPCESM200	Earth and Physical Systems	MPC	4	5	3	2	0

L — Lecture, P — Practical/Practicum , O — On the Job Training

Semester: 4

Course Code	Title of the Course	Type of the Course	Credit	Hours /week	Hour Distribution /week		
					L	P	O
MG4MPCESM200	Environmental Pollution and Management	MPC	4	5	3	2	0

L — Lecture, P — Practical/Practicum, O — On the Job Training

Semester: 5

Course Code	Title of the Course	Type of the Course	Credit	Hours/ week	Hour Distribution /week		
					L	P	O
MG5MPCESM300	Environmental Disasters	MPC	4	4	4	0	0

L — Lecture, P — Practical/Practicum , O — On the Job Trainin

Semester: 6

Course Code	Title of the Course	Type of the Course	Credit	Hours/ week	Hour Distribution /week		
					L	P	O
MG6MPCESM300	Natural Resource Management	MPC	4	4	4	0	0

L — Lecture, P — Practical/Practicum , O — On the Job Training

Syllabus



SEMESTER 1

MGU-B.VOC. (HONOURS)

Syllabus



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Science				
Programme					
Course Name	Environment and Ecology				
Type of Course	MPC				
Course Code	MG1MPCESM100				
Course Level	100				
Course Summary	This course provides students with an understanding on basics in environment science and its types and components. Throughout the course, students will engage in discussions aimed at understanding the intricate relationships between living organisms and their environment, and the importance of sustainable practices to protect our planet's ecosystems.				
Semester	1	Credits		4	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
		3	1	0	75
Pre- requisites, if any	Nil				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains *	PO No
1	Understand the term Environment and its components	K, U	3,6
2	Understand the principles, scope and of ecology	K, U	2, 3
3	Compare the structure, function and characteristics of different types of ecosystems.	U	3,6
4	Analyze the trophic relations and energy flow in an ecosystem.	U, An	1
5	Analyze the various population interactions in an ecosystem.	U, An	1,3
6	Describe gradients and diversity measures in community	K, U	2
*Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap) }			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	0	0	2	0	0	2	0	0	0	0
CO 2	0	2	2	0	0	0	0	0	0	0
CO 3	0	0	3	0	0	2	0	0	0	0
CO 4	2	0	0	0	0	0	0	0	0	0
CO 5	2	0	2	0	0	0	0	0	0	0
CO 6	0	2	0	0	0	0	0	0	0	0

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level)

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Environmental Ecology		10 hrs	
	1.1	Environment: Definition, types and components of Environment	2	1
	1.2	Ecology: Definitions and Subdivisions of ecology - autecology and synecology	2	2
	1.3	Scope of Ecology	2	2
	1.4	Social Ecology: Deep Ecology and Shallow ecology	2	2
	1.5	Ecological Terminologies	2	1,2
2	Ecosystem		16 hrs	
	2.1	Ecosystems Definition, components, Structure and Functions and types of ecosystem	3	3
	2.2	Food Chain and Food Web Energy Flow in an Ecosystem	2	4
	2.3	Major Ecosystems: Forest Ecosystem Grassland Ecosystem, Desert Ecosystem	3	3
	2.4	Freshwater and marine ecosystem Wetland ecosystem, mangrove ecosystem, Estuarine ecosystem	3	3
	2.7	Practical- Demonstration of Food web of any of the ecosystem(aquatic/ Terrestrial	5	3,4

3	Population Ecology			9 hrs	
	3.1	Definition, Characteristics of Population: Density, Growth rate, Natality, Mortality and carrying capacity	3	5	
	3.2	Population distribution patterns: Uniform, Random and Clumped	2	5	
	3.3	Population Interactions: competition, Predation, mutualism, commensalism, parasitism and allelopathy.	4	5	
4	Community Ecology			40 hrs	
	4.1	Definition, Classification and Characteristics of Community: Species diversity, Species richness, Dominance, Abundance	5	6	
	4.2	Community gradients: Ecotone and Edge Effect	2	6	
	4.3	Ecological Niche	4	3,6	
	4.3	Keystone species and Dominant species	4	3,6	
	4.4	Practical: Community study quadrat method; flora and fauna study by Frequency, Density and Abundance Field trip to biodiversity hotspot: Organize a visit to a local biodiversity- rich area (e.g., national park, wetland, or protected reserve) to observe and document various species, ecosystems, and their interdependencies	25	6	
5	Teacher specific content				

MGU-B.VOC. (HONOURS)

Syllabus

Teaching and Learning	Classroom Procedure (Mode of transaction) Lectures, Demonstration, Animations, Presentations, Discussions.																				
Assessment Types	MODE OF ASSESSMENT A. Continuous Comprehensive Assessment (CCA) Theory <table border="1"> <tr> <td colspan="2">Total Mark: 25</td></tr> <tr> <td colspan="2">Assessment methods</td></tr> <tr> <td>Assignment</td><td>10</td></tr> <tr> <td>Seminar/ Quiz/ Group Discussion</td><td>5</td></tr> <tr> <td>Test</td><td>10</td></tr> </table> Practical <table border="1"> <tr> <td colspan="2">Total Mark: 15</td></tr> <tr> <td colspan="2">Assessment methods</td></tr> <tr> <td>Involvement</td><td>5</td></tr> <tr> <td>Punctuality</td><td>5</td></tr> <tr> <td>Record</td><td>5</td></tr> </table>	Total Mark: 25		Assessment methods		Assignment	10	Seminar/ Quiz/ Group Discussion	5	Test	10	Total Mark: 15		Assessment methods		Involvement	5	Punctuality	5	Record	5
Total Mark: 25																					
Assessment methods																					
Assignment	10																				
Seminar/ Quiz/ Group Discussion	5																				
Test	10																				
Total Mark: 15																					
Assessment methods																					
Involvement	5																				
Punctuality	5																				
Record	5																				

B. End Semester Evaluation (ESE) Theory

Total mark: 50		
Assessment methods: Written Exam		
Duration of Examination: 1.5 hrs		
Pattern of Examination: Non-MCQ		
Part A	1 mark	Answer any 15 out of 17
Part B	5 mark	Answer any 3 out of 5
Part C	10 mark	Answer any 2 out of 4

Part A can be objective type, fill in the blanks, multiple choice etc.

Practical

Total mark: 35	
Duration of Examination: 2 hrs	
Assessment methods	
Theory/ Procedure	10
Skill and Performance	10
Calculation and Result	10
Viva	5

References

1. Abbasi, S. A. (1998). *Environmental pollution and its control*. Cogent International.
2. Abbasi, S. A. (1998). *Water quality: Sampling and analysis*. Discovery Publishing House.
3. Agarwal, K. C. (1996). *Biodiversity*. Agrobotanical Publishers.
4. Agarwal, K. C. (1993). *Environmental biology* (2nd ed.). Agrobotanical Publishers.
5. APHA. (1998). *Standard methods for the examination of water and wastewater*. Moduleed Book Press.
6. Sharma, J. P. (2009). *Environment and ecology*. University Science Press.



SEMESTER 2

MGU-B.VOC. (HONOURS)

Syllabus



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Science			
Programme				
Course Name	Environmental Education and Biodiversity Conservation			
Type of Course	MPC			
Course Code	MG2MPCESM100			
Course Level	100			
Course Summary	This course provides students with an understanding about the needs of environmental education and the role of peoples and NGOs in environmental protection. This course allows the students to learn the fundamentals of biodiversity and conservation biology. Students will also get the knowledge and skills required to conserve biodiversity and prevent the extinction of species in the face of rapid increases in human impacts on the planet.			
Semester	2	Credits		Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT
		4	0	0
Pre-requisites, if any	NIL			

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains	PO No
1	Describe the scope and importance of ecology and environmental science	R, U	1
2	Build central ideas of philosophical approaches in environmental ethics	E	1, 2
3	Build central ideas behind human Ecology, including knowledge of the history, Conservation Movements, People and Nature, Indigenous communities	E	5
4	Discuss and examine the values and threats affecting biodiversity	C, An, E	4
5	Understand and evaluate the various initiatives for biodiversity conservation.	U,E	7, 10
*Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap) }			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	0	0	0	0	0	0	0	0	0
CO 2	3	2	0	0	0	0	0	0	0	0
CO 3	0	0	0	0	3	0	0	0	0	0
CO 4	0	0	0	3	0	0	0	0	0	0
CO 5	0	0	0	0	0	0	2	0	0	2

‘0’ is No Correlation, ‘1’ is Slight Correlation (Low level), ‘2’ is Moderate Correlation (Medium level) and ‘3’ is Substantial Correlation (High level).

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Introduction to Environmental Studies		4 hrs	
	1.1	Environmental Studies: Scope, Importance	2	1
	1.2	Multidisciplinary nature of environmental Studies	2	1,2
2	Environmental Ethics and Education		12 hrs	
	2.1	Environmental Ethics	2	2
	2.2	Environmentalism, Biocentrism, Ecocentrism and Anthropocentrism	4	2
	2.3	Environmental Education: Goals, Objectives and Principles,	4	1
	2.4	Environmental Education: Scope and Importance	2	1,2
3	Environmental movement, Participation and Awareness		21 hrs	
	3.1	Need and objectives for Public Awareness	2	3
	3.2	Environmental movement and Peoples participation	4	3
	3.3	Well Known Environmental Movements in India	5	3,4
	3.4	Role of Tribal people, Women and NGOs in Environmental Protection	5	3,4,5
	3.5	Various national and international NGOs	5	3,4,5
4	Biodiversity and Conservation		23 hrs	
	4.1	Biodiversity: Introduction, definition, types of biodiversity - species, genetic and ecosystem diversity.	3	3
	4.2	Benefits of Biodiversity	3	3,4
	4.3	Threats to Biodiversity	4	3,4
	4.4	Biodiversity Hotspots	4	3,5
	4.5	IUCN Red list categories	3	3,4,5
	4.6	Biodiversity Conservation Strategies: Insitu and Exsitu	6	3,4,5
5	Teacher specific content			

Teaching and Learning Approach	Classroom Procedure (Mode of transaction) Leverage a blended learning approach with a mix of lectures, interactive discussions																														
Assessment Types	MODE OF ASSESSMENT																														
	A. Continuous Comprehensive Assessment (CCA) Theory <table border="1"><tr><td colspan="2">Total Mark: 30</td></tr><tr><td colspan="2">Assessment methods</td></tr><tr><td>Assignment</td><td>10</td></tr><tr><td>Seminar/ Quiz/ Group Discussion</td><td>10</td></tr><tr><td>Test</td><td>10</td></tr></table> B. End Semester Evaluation (ESE) Theory <table border="1"><tr><td colspan="3">Total mark: 70</td></tr><tr><td colspan="3">Assessment methods: Written Exam</td></tr><tr><td colspan="3">Duration of Examination: 2 hrs</td></tr><tr><td colspan="3">Pattern of Examination: Non-MCQ</td></tr><tr><td>Part A</td><td>1 mark</td><td>Answer any 25 out of 27</td></tr><tr><td>Part B</td><td>5 mark</td><td>Answer any 5 out of 7</td></tr><tr><td>Part C</td><td>10 mark</td><td>Answer any 2 out of 4</td></tr></table> Part A can be objective type, fill in the blanks, multiple choice etc.	Total Mark: 30		Assessment methods		Assignment	10	Seminar/ Quiz/ Group Discussion	10	Test	10	Total mark: 70			Assessment methods: Written Exam			Duration of Examination: 2 hrs			Pattern of Examination: Non-MCQ			Part A	1 mark	Answer any 25 out of 27	Part B	5 mark	Answer any 5 out of 7	Part C	10 mark
Total Mark: 30																															
Assessment methods																															
Assignment	10																														
Seminar/ Quiz/ Group Discussion	10																														
Test	10																														
Total mark: 70																															
Assessment methods: Written Exam																															
Duration of Examination: 2 hrs																															
Pattern of Examination: Non-MCQ																															
Part A	1 mark	Answer any 25 out of 27																													
Part B	5 mark	Answer any 5 out of 7																													
Part C	10 mark	Answer any 2 out of 4																													

References

1. Gaur, S., & Chandrasekhar, T. (n.d.). *Environmental management: Role of NGOs, risk perception and its mitigation*.
2. Misra, S. P., & Pandey, S. N. (n.d.). *Essential environmental studies*. Ane Books Pvt. Ltd.
3. Sharma, J. P. (n.d.). *Environment and ecology*. University Science Press.
4. Sharma, V. S. (n.d.). *Environmental education*. New Central Publications.
5. Santra, S. C. (n.d.). *Environmental science*. New Central Book Agency Pvt. Ltd.
6. Raghavan, S. (n.d.). *India's environmental policy with special reference to marine environment*.
7. Sundar, I. (n.d.). *Environment and sustainable development*.



MGU-B.VOC. (HONOURS)

Syllabus



SEMESTER 3

MGU-B.VOC. (HONOURS)

Syllabus



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Science				
Programme					
Course Name	Earth and Physical Systems				
Type of Course	MPC				
Course Code	MG3MPCESM200				
Course Level	200				
Course Summary	This course is designed to provide a comprehensive understanding of Earth and the physical system.				
Semester	3	Credits		4	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
		3	1	0	
Pre- requisites, if any	Nil				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains *	PO No
1	Understand basics of Earth and its various physical systems	U	3
2	Describe the composition of different physical systems.	R	4
3	Examine how these physical systems interact	An	2
4	Discuss weather and climate	C	3
5	Experiment with weather monitoring equipment	A	1,10
6	Understanding the basics of Oceanography	U	3
*Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap) }			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	0	0	3	0	0	0	0	0	0	0
CO 2	0	0	0	3	0	0	0	0	0	0
CO 3	0	3	0	0	0	0	0	0	0	0
CO 4	0	0	3	0	0	0	0	0	0	0
CO 5	2	0	0	0	0	0	0	0	0	3
CO 6	0	0	3	0	0	0	0	0	0	0

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level)

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Introduction to Earth and its composition		12 hrs	
	1.1	The Earth in relation to the Universe-Origin of the earth	2	1
	1.2	Structure and composition of interior of the earth	4	1,2
	1.3	Representation of earth: Globes & Maps	2	1
	1.4	Parallels and Meridians	2	1
	1.5	Biosphere and its divisions-Atmosphere-Lithosphere-Hydrosphere	2	1,2,3
2	Atmosphere		45 hrs	
	2.1	Atmosphere: Composition and Vertical Stratification	3	1,2
	2.2	Wind- Definition and classification. Pressure belts	6	1,2,3
	2.3	Definition of Weather and climate. Familiarization of weather monitoring equipment	2	4,
	2.4	Indian Monsoon: onset of Monsoon and retreat of monsoon, rain bearing systems, Break in the monsoon	4	1,2,4
	2.7	Practical- Field visit to weather monitoring station. Meteorological data collection and analysis (Rainfall, Evaporation, Temperature, Relative humidity, wind speed, wind direction, Wind-rose etc)	30	5

3	Lithosphere			10hrs	
	3.1	Rocks- Classification-igneous-		2	1
		sedimentary –metamorphic;			
	3.2	Soil- Formation; Soil profile and basic classification		2	2
	3..3	Soil types of India and Kerala		6	1
4	Hydrosphere			8 hrs	
	4.1	Hydrology; Types of water- The surface water and ground water. Their relationship		2	1,3
	4.2	Ground water-origin and classification. Water table		2	1,2
	4.3	Hydrological cycle: Drainage basins, catchments and watershed		2	1,3
	4.4	Introduction to Oceanography		2	5
5	Teacher specific content				

Teaching and Learning Approach	Classroom Procedure (Mode of transaction) Lectures, Demonstration, Animations, Presentations, Discussions.											
Assessment Types	MODE OF ASSESSMENT											
	A. Continuous Comprehensive Assessment (CCA)											
	Theory											
	<table><tr><td colspan="2">Total Mark: 25</td></tr><tr><td colspan="2">Assessment methods</td></tr><tr><td>Assignment</td><td>10</td></tr><tr><td>Seminar/ Quiz/ Group Discussion</td><td>5</td></tr><tr><td>Test</td><td>10</td></tr></table>		Total Mark: 25		Assessment methods		Assignment	10	Seminar/ Quiz/ Group Discussion	5	Test	10
	Total Mark: 25											
	Assessment methods											
Assignment	10											
Seminar/ Quiz/ Group Discussion	5											
Test	10											
Practical												
<table><tr><td colspan="2">Total Mark: 15</td></tr><tr><td colspan="2">Assessment methods</td></tr><tr><td>Involvement</td><td>5</td></tr><tr><td>Punctuality</td><td>5</td></tr><tr><td>Record</td><td>5</td></tr></table>		Total Mark: 15		Assessment methods		Involvement	5	Punctuality	5	Record	5	
Total Mark: 15												
Assessment methods												
Involvement	5											
Punctuality	5											
Record	5											

B. End Semester Evaluation (ESE) Theory		
Total mark: 50		
Assessment methods: Written Exam		
Duration of Examination: 1.5 hrs		
Pattern of Examination: Non-MCQ		
Part A	1 mark	Answer any 15 out of 17
Part B	5 mark	Answer any 3 out of 5
Part C	10 mark	Answer any 2 out of 4
Part A can be objective type, fill in the blanks, multiple choice etc.		
Practical		
Total mark: 35		
Duration of Examination: 2 hrs		
Assessment methods		
Theory/ Procedure	10	
Skill and Performance	10	
Calculation and Result	10	
Viva	5	

References

1. Misra, S. P., & Pandey, S. N. (2010). *Essential environmental studies*. Ane Books Pvt. Ltd.
2. Holmes, A. (1965). *Principles of physical geology*. FLBS.
3. Barry, R. G., & Chorley, R. J. (1998). *Atmosphere, weather and climate* (7th ed.). Routledge.
4. Siddhartha, K. (2005). *Atmosphere, weather and climate*. Kisalaya Publications Pvt. Ltd.
5. Das, P. K. (1995). *The monsoons* (3rd ed.). National Book Trust India.
6. Siddhartha, K. (1999). *Oceanography: A brief introduction*. Kisalaya Publications Pvt. Ltd.



SEMESTER 4

MGU-B.VOC. (HONOURS)

Syllabus



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Science				
Programme					
Course Name	Environmental Pollution and Management				
Type of Course	MPC				
Course Code	MG4MPCESM200				
Course Level	200				
Course Summary	The course aims to equip students with the knowledge and skills necessary to understand, assess, manage, and mitigate environmental pollution while considering sustainable and socially responsible practices.				
Semester	4	Credits		4	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
		3	1	0	75
Pre- requisites, if any	Nil				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains *	PO No
1	To build an idea on various types of pollutions.	U	1
2	Recognize the sources and types of water, air, and soil pollution.	R	3,10
3	Discuss the Fate and effects of pollutants	C	1
4	Explain the types and sources of Noise and Radioactive pollution	E	3,10
*Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap) }			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	0	0	0	0	0	0	0	0	0
CO 2	0	0	3	0	0	0	0	0	0	3
CO 3	3	0	0	0	0	0	0	0	0	0
CO 4	0	0	3	0	0	0	0	0	0	2

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level)

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Air pollution - Sources, Effects.		12 hrs	
	1.1	Air pollution - Sources, effects and control measures Definition - Structure and constituents; sources, nature and types of pollutants - primary and secondary air pollutant, particulate matter, aerosols, fly ash, bio pollutants.	8	1, 2
	1.2	Effects of air pollution on - Human being - Animals - Plants - Materials, buildings and climate.	4	3
2	Water pollution - Sources and Effects		30 hrs	
	2.1	Definition, Types of water pollution - Point and non-point source of water pollution- surface and ground water pollution.	3	1
	2.2	Sources of water pollution - Domestic, Industrial, Agricultural and Natural sources.	2	2
	2.3	Effects of water pollution on human beings, animals, plants and environment.	5	3
		1. Determination of pH, EC, Turbidity 2. Estimation of Total Solids, Total Suspended Solids, Total Dissolved Solids	20	3
3	Soil / Land pollution - Sources, Effects		20 hrs	
	3.1	Sources, nature and types of soil pollutants	5	1,2
	3.2	Effects of soil pollution on Environment	5	3
	3.3	1. Determination of pH in soil sample 2. Moisture content in soil sample	10	3

4	Noise pollution and Radioactive pollution - Definition, Effects			13hrs	
	4.1	Definition, sources, general effects of noise pollution.	5	4	
	4.2	Definition, Types and sources of Radioactivity.	4	4	
	4.3	Effects of Radioactive pollution	4	4	
5	Teacher specific content				

Teaching and Learning Approach	Classroom Procedure (Mode of transaction) Lectures, Demonstration, Animations, Presentations, Discussions.
Assessment Types	MODE OF ASSESSMENT
	A. Continuous Comprehensive Assessment (CCA) Theory
	Total Mark: 25
	Assessment methods
	Assignment10
	Seminar/ Quiz/ Group Discussion5
	Test10
	Practical
	Total Mark: 15
	Assessment methods
Involvement5	
Punctuality5	
Record5	

**B. End Semester Evaluation (ESE)
Theory**

Total mark: 50		
Assessment methods: Written Exam		
Duration of Examination: 1.5 hrs		
Pattern of Examination: Non-MCQ		
Part A	1 mark	Answer any 15 out of 17
Part B	5 mark	Answer any 3 out of 5
Part C	10 mark	Answer any 2 out of 4

Part A can be objective type, fill in the blanks, multiple choice etc.

Practical

Total mark: 35	
Duration of Examination: 2 hrs	
Assessment methods	
Theory/ Procedure	10
Skill and Performance	10
Calculation and Result	10
Viva	5

References

1. Abbasi, S. A. (1998). *Environmental pollution and its control*. Coent International.
2. Abbasi, S. A. (1998). *Water quality: Sampling and analysis*. Discovery Publishing House.
3. Pani, B. (n.d.). *Textbook of environmental chemistry and pollution control*. I. K. International Publishing House Pvt. Ltd.
4. Dara, S. S. (n.d.). *A textbook of environmental chemistry and pollution control*. S. Chand and Company Ltd.



SEMESTER 5

MGU-B.VOC. (HONOURS)

Syllabus



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Science				
Programme					
Course Name	Environmental Disasters				
Type of Course	MPC				
Course Code	MG5MPCESM300				
Course Level	300				
Course Summary	The course deals with the major natural and anthropogenic disasters its environmental constraints. The course also explains on the basic relation between disasters and development.				
Semester	5	Credits		4	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
		4	0	0	60
Pre- requisites, if any	Nil				

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains *	PO No
1	To provide basic conceptual understanding of disasters.	U	1, 10
2	Explain about concepts of disaster management and types of disasters	C	10
3	To understand approaches of Disaster Management	An	3
4	To build skills to respond to disaster	S	1
*Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap) }			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	3	0	0	0	0	0	0	0	0	3
CO 2	0	0	0	0	0	0	0	0	0	3
CO 3	0	0	3	0	0	0	0	0	0	0
CO 4	3	0	0	0	0	0	0	0	0	0

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level)

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Environment and Disasters		10hrs	
	1.1	Science and Facts of Natural Hazards. Earth's processes as disasters: Internal and external	5	1
	1.2	Climate change and Disasters	5	1
2	Types and Classification of Disasters		16hrs	
	2.1	Natural Disasters: Meteorological disasters, Geological disasters, Biological disasters	8	2
	2.2	Anthropogenic Disasters: Chemical, Industrial and Nuclear related Disasters, Accident related Disasters	8	2
3	Disaster Management Concepts		20 hrs	
	3.1	Introduction to key concepts, terminologies and their complexities (Hazard, vulnerability, Exposure, Risk, Crisis, emergencies, Vulnerability, Disasters, Resilience)	5	1
	3.2	Community based Disaster Management	15	3
4	Disaster Management in Indian Context		14 hrs	
	4.1	Major Disasters in India	8	1
	4.2	Challenges of disasters in India	6	4
5	Teacher specific content			

Teaching and Learning Approach	Classroom Procedure (Mode of transaction) Lectures, Demonstration, Presentations, Discussions.																				
Assessment Types	MODE OF ASSESSMENT																				
	A. Continuous Comprehensive Assessment (CCA) Theory <table><tr><td colspan="3">Total Mark: 30</td></tr><tr><td colspan="3">Assessment methods</td></tr><tr><td>Assignment</td><td></td><td>10</td></tr><tr><td>Seminar/ Quiz/ Group Discussion</td><td></td><td>10</td></tr><tr><td>Test</td><td></td><td>10</td></tr></table>	Total Mark: 30			Assessment methods			Assignment		10	Seminar/ Quiz/ Group Discussion		10	Test		10					
	Total Mark: 30																				
Assessment methods																					
Assignment		10																			
Seminar/ Quiz/ Group Discussion		10																			
Test		10																			
B. End Semester Evaluation (ESE) Theory <table><tr><td colspan="3">Total mark: 70</td></tr><tr><td colspan="3">Assessment methods: Written Exam</td></tr><tr><td colspan="3">Duration of Examination: 2 hrs</td></tr><tr><td colspan="3">Pattern of Examination: Non-MCQ</td></tr><tr><td>Part A</td><td>1 mark</td><td>Answer any 25 out of 27</td></tr><tr><td>Part B</td><td>5 mark</td><td>Answer any 5 out of 7</td></tr><tr><td>Part C</td><td>10 mark</td><td>Answer any 2 out of 4</td></tr></table>	Total mark: 70			Assessment methods: Written Exam			Duration of Examination: 2 hrs			Pattern of Examination: Non-MCQ			Part A	1 mark	Answer any 25 out of 27	Part B	5 mark	Answer any 5 out of 7	Part C	10 mark	Answer any 2 out of 4
Total mark: 70																					
Assessment methods: Written Exam																					
Duration of Examination: 2 hrs																					
Pattern of Examination: Non-MCQ																					
Part A	1 mark	Answer any 25 out of 27																			
Part B	5 mark	Answer any 5 out of 7																			
Part C	10 mark	Answer any 2 out of 4																			
	Part A can be objective type, fill in the blanks, multiple choice etc.																				

References

1. Government of India–UNDP Disaster Risk Program. (2009–2012). *Disaster management guidelines*.
2. Coppola, D. P. (2006). *Introduction to international disaster management*. Butterworth-Heinemann.
3. Gupta, A. K., Niar, S. S., & Chatterjee, S. (2013). *Disaster management and risk reduction: Role of environmental knowledge*. Narosa Publishing House.
4. Murthy, D. B. N. (2012). *Disaster management*. Deep and Deep Publication Pvt. Ltd.



SEMESTER 6

MGU-B.VOC. (HONOURS)

Syllabus



Mahatma Gandhi University Kottayam

Faculty/ Discipline	Science				
Programme					
Course Name	Natural Resource Management				
Type of Course	MPC				
Course Code	MG6MPCESM300				
Course Level	300				
Course Summary	This course allows the students to learn the fundamentals theories and concepts of Environment Management.				
Semester	6	Credits		4	Total Hours
Course Details	Learning Approach	Lecture	Practical	OJT	
		4	0	0	60
Pre- requisites, if any	Nil				

Syllabus

COURSE OUTCOMES (CO)

CO No.	Expected Course Outcome	Learning Domains *	PO No
1	Understand the basic concepts of Natural Resource Management	U	7
2	Evaluate and apply fundamental concepts of land and water resource management	E, A	3
3	Analyze the management of various mineral resources	An	1
4	Apply participatory approaches in community based natural resource management	A,C	4,8
*Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap) }			

CO-PO ARTICULATION MATRIX

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	0	0	0	0	0	0	3	0	0	0
CO 2	0	0	3	0	0	0	0	0	0	0
CO 3	3	0	0	0	0	0	0	0	0	0
CO 4	0	0	0	3	0	0	0	3	0	0

'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level)

COURSE CONTENT

Content for Classroom transaction (Units)

Module	Units	Course description	Hrs	CO No.
1	Natural Resources		10 hrs	
	1.1	Concept and types of Natural Resources	2	1
	1.2	Depletion of Natural Resources	4	1
	1.3	Conservation of Natural Resources	4	1,4
2	Management of Physical Resources		16 hrs	
	2.1	Land resources	2	2
	2.2	Land degradation and its causes	2	2
	2.3	Desertification and its control	2	2
	2.4	Soil conservation strategies	3	2
	2.5	Wastelands and their reclamation	3	2
	2.6	Sustainable agriculture	4	2,4
3	Management of Minerals		16	
	3.1	Types of mineral resources	2	3
	3.2	Mineral wealth of India	4	3
	3.3	Environmental impacts of mineral exploration	6	3
	3.4	Conservation of minerals	4	3,4
4	Management of Water Resources		18 hrs	
	4.1	Types of water resources	2	2
	4.2	Water conservation and conservation strategies	6	2
	4.3	Surface and ground water resources with special reference to Kerala	4	2
	4.4	Introduction to integrated water resources	6	2,4
5	Teacher specific content			

Teaching and Learning	Classroom Procedure (Mode of transaction) Lectures, Demonstration, Animations, Presentations, Discussions.																				
Assessment Types	MODE OF ASSESSMENT																				
	A. Continuous Comprehensive Assessment (CCA) Theory <table><tr><td colspan="2">Total Mark: 30</td></tr><tr><td colspan="2">Assessment methods</td></tr><tr><td>Assignment</td><td>10</td></tr><tr><td>Seminar/ Quiz/ Group Discussion</td><td>10</td></tr><tr><td>Test</td><td>10</td></tr></table>	Total Mark: 30		Assessment methods		Assignment	10	Seminar/ Quiz/ Group Discussion	10	Test	10										
	Total Mark: 30																				
Assessment methods																					
Assignment	10																				
Seminar/ Quiz/ Group Discussion	10																				
Test	10																				
B. End Semester Evaluation (ESE) Theory <table><tr><td colspan="3">Total mark: 70</td></tr><tr><td colspan="3">Assessment methods: Written Exam</td></tr><tr><td colspan="3">Duration of Examination: 2 hrs</td></tr><tr><td colspan="3">Pattern of Examination: Non-MCQ</td></tr><tr><td>Part A</td><td>1 mark</td><td>Answer any 25 out of 27</td></tr><tr><td>Part B</td><td>5 mark</td><td>Answer any 5 out of 7</td></tr><tr><td>Part C</td><td>10 mark</td><td>Answer any 2 out of 4</td></tr></table>	Total mark: 70			Assessment methods: Written Exam			Duration of Examination: 2 hrs			Pattern of Examination: Non-MCQ			Part A	1 mark	Answer any 25 out of 27	Part B	5 mark	Answer any 5 out of 7	Part C	10 mark	Answer any 2 out of 4
Total mark: 70																					
Assessment methods: Written Exam																					
Duration of Examination: 2 hrs																					
Pattern of Examination: Non-MCQ																					
Part A	1 mark	Answer any 25 out of 27																			
Part B	5 mark	Answer any 5 out of 7																			
Part C	10 mark	Answer any 2 out of 4																			

Part A can be objective type, fill in the blanks, multiple choice etc.

References

1. Santra, S. C. (2014). *Environmental science*. New Central Book Agency Pvt. Ltd.
2. Gangstad, I. (1990). *Natural resource management of water and land*. Van Norstrand.
3. Abbasi, S. A. (2001). *Water resources projects and their environmental impacts*. Discovery Publishing House.



MGU-B.VOC. (HONOURS)

Syllabus